

Numerical Method

CSIT — 3rd Semester — 2082 — Answer Sheet
Subject Code: CSC212

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. A software project manager wants to study the relationship between the number of lines of code (LOC) written and the number of bugs reported during testing. The data from several modules is shown below.

Lines of Code (LOC)	Bugs Reported
100	15
200	25
300	30
691	13
171	7

Fit a straight line to model the relationship between LOC and bugs reported.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Some observation records the displacement of a particle at different time intervals to study its motion. The data collected is given below.

Time (t in sec)	Displacement (s in m)
2.5	3.0
3.5	5.1
5.4	10.5
7.4	20.9
10.5	37.0

Construct the divided difference table for the displacement data and estimate the particle's velocity and acceleration at $t = 2.5$ seconds.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Solve the following system of linear equations using the Jacobi iteration method.

$$2x + 3y - z = 14$$
$$x - y + 2z = 2$$
$$-3x + 2y + 3z = 3$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Explain how boundary value problem differs from initial value problem. Explain how shooting method is used to solve boundary value problem.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Find the approximate value of y when $x = 0.4$ for the differential equation $dy/dx = x^2 + y$, given that $y = 2$ when $x = 0$, using the Runge-Kutta method of order 4 with a step size of $h = 0.2$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Consider a metallic plate of size 36cm × 36cm. If two of the sides are held at 50 degree celsius and the other two sides at 0 degrees celsius, find the steady state temperatures of interior points, assuming a grid size of 12cm × 12cm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Derive the secant method for solving non-linear equation and using this method solve $3x^3 - x - 5 = 0$, correct up to three decimal places.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. How does pivoting improve accuracy of solution? Explain how Gauss elimination method differs from Gauss Jordan method. Solve the following system of linear equations using Gauss elimination with partial pivoting: $x + 2y - z = 13$, $x - y + 2z = 122$, $x + y + 2z = 10$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is numerical integration and why is numerical integration used instead of analytical integration? Write an algorithm and a program to compute the integration using Simpson's 1/3 rule.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Discuss the sources of errors in numerical calculations and explain how each type affects computational results.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. How regression differ from interpolation? Construct the cubic spline for the following data points. X123Y235

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. A meteorologist is tracking the temperature at different times during the day to predict the temperature at a specific hour. The following temperature data (in °C) was recorded at different hours: Time (t)(hrs) 5 8 10 13 Temperature (T)(°C) 18 22 24 26 Estimate the temperature at $t = 9$ using Lagrange's interpolation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.